Elena Alegret Regalado

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Profile

AI/ML researcher in 3D computer vision and generative models, **pursuing an M.Sc. at ETH Zürich**. Four research internships and **two papers at 3DV 2026** (one as co-first author), in collaboration with researchers from Google, Microsoft, ETH and TUM.

Research Interests: 3D Generative Models, 3D Computer Vision, 3D Reconstruction and Understanding, Segmentation, Robotics, Autonomous Driving, Healthcare applications.

Education

2025 – Present **M.Sc. in Electrical Engineering and Information Technology** ***ETH Zürich***
Major: Signal Processing and Machine Learning. Expected graduation: **2028**
Relevant courses: 3D Vision, Deep Learning, Probabilistic Artificial Intelligence

2024 – 2025 **Visiting Student** ***Technical University of Munich (TUM), Germany***
One-year exchange. GPA: 10/10. Bachelor's thesis graded **Highest Honors**.

2021 – 2025 **B.Sc. in Artificial Intelligence** ***Universitat Politècnica de Catalunya (UPC), Spain***
Top 10 of the class, GPA: 8.55/10.
1st Class Honors: Voice and Dialog Processing, Advanced Models, HPC, Ethics.

Publications

1. **E. Alegret***, K. Li*, S. Wang, S. Liang, M. Niemeyer, S. Gasperini, N. Navab, F. Tombari. *GALA: Guided Attention with Language Alignment for Open-Vocabulary Gaussian Splatting.* International Conference on 3D Vision (**3DV 2026**). arXiv, Project page.

2. S. Wang, K. Li, S. Liang, **E. Alegret**, J. Ma, N. Navab, S. Gasperini. *Visibility-Aware Language Aggregation for Open-Vocabulary Segmentation in 3D Gaussian Splatting.* International Conference on 3D Vision (**3DV 2026**). arXiv.

* Equal contribution.

Research Experience

Feb 2026 – Present **Student Research Intern** ***ETH Zürich, CVG Lab***
Advisors: Prof. Federico Tombari, Prof. Marc Pollefeys, Daniel Béla Baráth, Michael Niemeyer and Kunyi Li

- Research on 3D generative models and neural rendering. Co-leading the development of a latent flow-matching model for high-fidelity 3D Gaussian Splatting reconstruction from single-view images, trained at scale on object- and scene-level datasets; aiming for submission to CVPR.

Apr 2025 – Jan 2026 **Research Intern** ***TUM, CAMP Lab***
Advisors: Prof. Federico Tombari, Michael Niemeyer, Kunyi Li and Sen Wang

- Co-led the development of GALA (Bachelor's thesis), an open-vocabulary 3D scene understanding framework on 3D Gaussian Splatting that learns instance-level features without supervision and aligns them with language via cross-attention, enabling 2D/3D querying with a low memory footprint.
- Outperformed state-of-the-art methods in open-vocabulary segmentation on real-world benchmarks: highest 2D mIoU on LERF-OVS (55.5) and +6–12 mIoU on ScanNet-v2, with 3D queries 300× faster than LangSplat. **Co-first-author paper accepted at 3DV 2026**.

Jul 2024 – Sep 2024 **Research Intern** ***EPFL, LTS4 Lab***
Advisors: Prof. Pascal Frossard and Cédric Vincent-Cuaz

- Built a fine-tuning and evaluation pipeline to benchmark state-of-the-art vision foundation models for cancer detection in X-ray images, and investigated graph-based representations for medical imaging.

Sep 2023 – Jun 2024 **Research Intern** ***Barcelona Supercomputing Center (BSC)***
Advisor: Prof. Dario García

- Developed Grad-CAM-guided de-biasing for medical-image classifiers and unified multiple medical-imaging datasets into one training pipeline.

Projects

Jun 2024 – Jul 2024 **AI-Driven Clothing Similarity** ***HackUPC***

- Built a deep learning model for visual similarity retrieval using Siamese/Triplet networks with semi-hard negative mining, enabling scalable look-alike and duplicate detection. Code: GitHub.

May 2024 – Jun 2024 **Explore and Find Node for Robot Exploration** ***UPC***

- Extended the ROS `frontier_exploration` node with YOLO-based object detection, creating a more robust and controlled autonomous exploration system. Code: GitHub.

Skills and Additional

Programming	Python, C++, MATLAB
ML & Tools	PyTorch, TensorFlow, Git, Docker, MLflow, ROS, LaTeX
Awards	Academic Excellence Award, Government of Catalonia (2021): top 3% in the university entrance exams
Hackathons	LauzHack (EPFL), HackTUM (TUM), HackUPC (UPC)
Courses	Stanford University — Convolutional Neural Networks for Visual Recognition (2022); BSC — NotOnlyFLOPs: Deep Learning Performance Optimization (2022) and Introduction to Docker Essentials (2024)
Languages	English (C1), Spanish (native), Catalan (native), French (B2), German (beginner)